

INFLECTION, INVERSION AND SUBJECT CLITICS

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"The inversion of order bringeth all to confusion."

-Thomas Fuller (1647), History of the Holy Warre

A number of languages contain processes which invert subjects and tensed elements. In this paper we shall build on earlier results of Safir (1980) and attempt to explain these phenomena, which may well be the source of Fuller's despairing remark. The questions we will ask are basic. Why do inversion rules exist? Why are they obligatory in some contexts and always ungrammatical in others? Why do a number of languages show constraints on tensed elements involving clausal second position (for example, the German "V/2" constraint). Why don't such constraints refer to some other position or to nodes other than those containing Tense?

This paper will briefly outline a simple answer to these questions, using material primarily from German and French. (See Safir (1980) for an analysis of English and some other languages.) In the first three sections of the paper, we will demonstrate how a few basic principles governing the behavior of "INFL", the node carrying tense and agreement features, can predict with surprising elegance the complex patterns of inversion found in our two languages. In the final section, we will present some speculations about why these principles exist and how they can be derived from more basic features of the theory of Universal Grammar.*

1. Principles

We generally assume the framework of the "Government- Binding" theory (see Chomsky (1981) and references therein). We rely particularly on a definition of government. We will suppose, with a number of researchers, that A is true:¹

A. INFL is the head of $\bar{S}$.

We will now give a working definition of government. Clause (i) is essentially the definition proposed by Aoun and Sportiche (in preparation), but differs minimally in empirical consequences from that of Chomsky (1981).

B. (i) α is governed by β^n if α and β^n are contained by $\beta^{\max}$ (perhaps, α and β^n adjacent).

(ii) Lexically filled COMP governs an adjacent constituent.

(iii) Trace is invisible for adjacency (unlike PRO); cf. Kayne (1980) on this point.

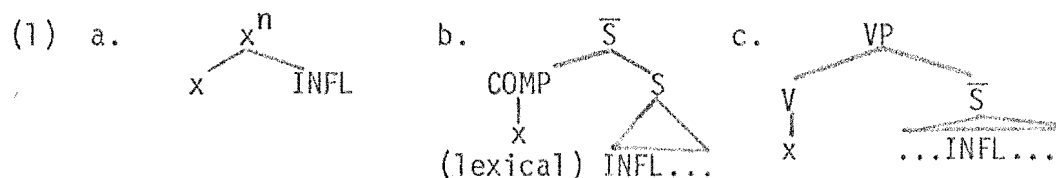
(iv) If a category is governed then its head is governed (cf. "percolation government", Kayne (1980)).

We now propose the following two principles:

C. INFL must be governed.

D. Trace of INFL must not be governed.

These principles will be derived in the last section. (1) below exemplifies our definition of government. In each case, INFL is governed:



In (1a), INFL is contained in a projection of x ; hence, by X-bar theory, it is contained by $x^{\max}$ and is governed by x . (1b) shows government from COMP. (1c) shows principle B(iv): $\bar{S}$ (the maximal projection of INFL) is governed by V, and INFL, its head, is governed accordingly.

To see how this system works in a simple case, we will consider the position of the inflected verb in German.

2. German

Let us suppose the following base rules for German. Note that we follow Thiersch (1978) and others in positing an underlying SOV order:

- (2) $\bar{S} \longrightarrow \text{COMP } S$
 $S \longrightarrow \text{NP } VP \text{ INFL}$

Finally we assume a D-structure (deep structure) rule which we may state informally as (3):

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(3) D-structure: INFL absorbs an adjacent V.

The nature of this rule is not of crucial importance here. We might derive its effects in various ways. Suppose, for example, that the node INFL in German is characterized by features non-distinct from V. Suppose also that German lacks independent lexical items which could be inserted into INFL to bear its features. If there were a constraint requiring that such features must have a lexical or referential carrier (cf. Lasnik (1980)), movement of V into INFL would be possible and effectively obligatory, much like do-support in some analyses of the English auxiliary.^{2,3} In any case, what is important is that our principles governing INFL will determine the position of the tensed verb in German. In particular, application of a general transformation like Move α , combined with landing site conventions such as those of Baltin (1978; forthcoming) will have the following effect:

(4) [_S ^V_{+INFL}] can move to [_S (Chomsky-adjunction).⁴

Let us consider matrix declarative sentences. The well-known generalization about German matrix declaratives is that one and only one x^{max} -constituent may precede the inflected verb. Following Koster (1975) and Thiersch (1978), we will suggest that this constituent has been moved to COMP. Thus:

- (5)a. [_S [_{COMP} gestern] [_S hat Hans das Buch dem Herrn gegeben]]
 'yesterday has Hans the+ACC book the+DAT man given'
 b. [_S [_{COMP} Hans] [_S hat das Buch dem Herrn gestern gegeben]]
 c. [_S [_{COMP} das Buch] [_S hat Hans dem Herrn gestern gegeben]]
 d. [_S [_{COMP} dem Herrn] [_S hat Hans das Buch gestern gegeben]]

Now note that our theory directly predicts this generalization. Consider the representation of (5a) if (4) has not applied:

- (6) [_S [_{COMP} gestern] [_S Hans [_{VP} das Buch dem Herrn gegeben] [_{INFL} ^{hat}]]

Examining the definition of government in B above, we find that hat, the inflected verb, is not governed. This violates C. (6) thus does not yield a well-formed surface string.

If, however, (4) does apply, and the inflected verb moves to a position adjacent to COMP, it will be governed under B(ii), since COMP is lexically filled. Notice that the filling of COMP, as in (5), is, correctly, obligatory. If COMP were not lexically filled, neither fronting of the verb nor any other operation would remedy the situation, since there would be no available governor for INFL. Note that this movement also satisfies principle D. A fuller representation of (5a) is (7):

- (7) [_S [_{COMP} gestern] [_S [_{INFL} ^{hat}]_i Hans das Buch dem Herrn gegeben _i]]

Note the trace left in the position vacated by the inflected verb. We already know that this position is ungoverned. This is what forced movement in the first place. Thus the trace is ungoverned, and this is precisely what is required by D. The interaction of Move α , principles C and D, and the theory of government thus yields the

V/2 generalization for German matrix declaratives.

Our analysis of declaratives will extend straightforwardly to matrix WH-questions, where WH-movement into COMP derives structures formally identical to (5-7). The inflected verb must move to a position adjacent to COMP, just as in (7), yielding output structures like:

- (8) [$\bar{S}$ [_{COMP} was] [_S [^{hat}_{INFL}]_i Hans dem Herrn gestern gegeben t_i]]
 'what has Hans the+DAT man yesterday given'

Our theory correctly predicts that the V/2 constraint will hold in WH-questions.⁵

Now consider the case of embedded declaratives. With certain marginal exceptions (see note 7), embedded declaratives, and indeed all embedded clauses, show obligatory verb-final order in German. This means that (4) must not apply. Embedded declaratives in German, as in English, occur as verbal objects and as complements of NPs, APs, etc. These are governed positions. Consider a structure like (9):

- (9) er hat [_{VP} mir gesagt [$\bar{S}$ dass [_S er geblieben [^{ist}_{INFL}]]]]⁶

The embedded clause is governed by the matrix verb gesagt. By B(iv) it follows that the INFL head of the embedded clause is also governed. C is satisfied.

Suppose we had applied (4) to (9):

- (10) er hat [_{VP} mir gesagt [$\bar{S}$ dass [_S [^{ist}_{INFL}]_i er geblieben t_i]]]]

The trace of the inflected verb remains in the base-generated head position of the embedded clause. We have seen, however, that this is a governed position. A trace in such a position violates D, and the structure is thus ruled out.

Embedded questions, once again, are formally identical to declaratives. They too have obligatory verb-final order. Thus, in (11a), the tensed verb of the subordinate clause is correctly governed, satisfying C. In (11b), however, this tensed verb has been fronted, leaving a governed trace and violating D:

- (11)a. Johann fragte [$\bar{S}$ wen [_S Fritz gesehen [^{habe}_{INFL}]]]
 'Johann asked who+ACC Fritz seen had'
 b. *Johann fragte [$\bar{S}$ wen [_S habe_i Fritz gesehen t_i]]

Thus, the theory of government and INFL sketched in the first section correctly captures the distinction between root sentences, where the V/2 constraint applies, and subordinate clauses, with obligatory verb-final order, without resort to special filters. It remains to show that our theory generalizes beyond German, and that it can be derived from other principles of universal grammar. This we do in the following sections.⁷

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3. French

French exhibits a complex set of inversions, with certain odd conditions, which have been discussed by Kayne (1972), Kayne & Pollock (1978) and others. We will show that our proposal, supplemented by a theory of subject clitics and nominative Case marking, accounts automatically for this range of data. In this sense, French is a real test of our theory.

We assume the following minimal base rules:

- (12) $\bar{S} \rightarrow \text{COMP } S$
 $S \rightarrow \text{NP INFL VP}$

INFL must be exempted from C in matrix declaratives, where COMP is empty. Suppose a special rule governs $\bar{S}$ in matrix declaratives; then INFL will always be governed by B(iv). This is necessary to allow (13):

- (13) Jean INFL est arrivé 'Jean has arrived'

It will not apply, in European Standard French, to matrix WH-questions, where COMP is filled:⁸

- (14) $*[\bar{S} \text{ quand } [S \text{ Jean INFL est arrivé}]]$ 'when...'

In (14), INFL is ungoverned, violating C.

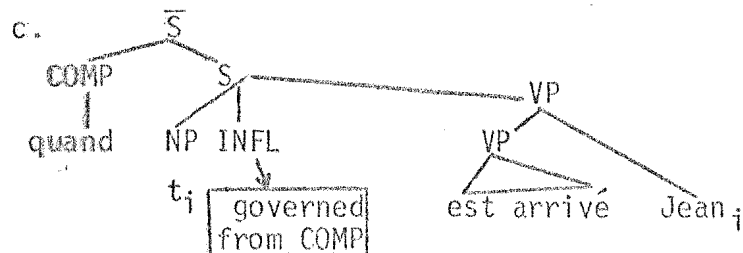
Functionally speaking, one might imagine two sorts of movement which would save a structure like (14): (a) Since trace is invisible for government by B(iii), one might move the subject NP, leaving INFL adjacent to the filled COMP and governed, by B(ii); (b) One might move INFL to a governed position, if its trace remained ungoverned. In fact, French realizes both strategies.

The first strategy is the familiar rule of Stylistic Inversion (Kayne & Pollock (1978)). As an exemplification of Move α , this process can be described as in (15):

- (15) NP can move to v_p] (Chomsky-adjunction)

This rule may apply in matrix or in embedded clauses:

- (16)a. Quand est arrivé Jean?
 b. Je ne sais pas quand est arrivé Jean.
 '(I don't know) when has arrived Jean'



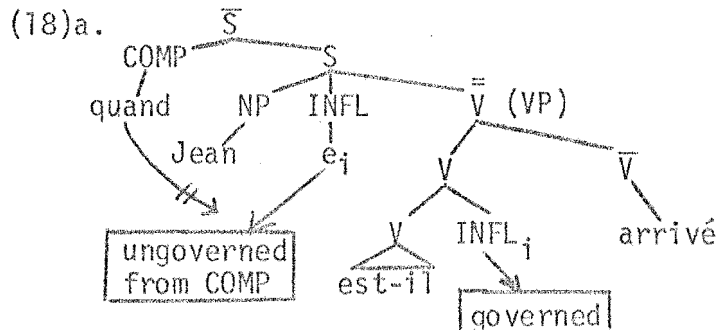
In (16c), the trace left in subject position does not block government of INFL by quand in COMP, and C is satisfied. Since INFL does not move, D is not invoked at all.

Note that in matrix sentence (16a) some process like Stylistic Inversion must apply to govern INFL. In (16b), however, INFL is already governed, by the matrix verb, under B(iv). In (16b), we correctly predict that inversion is optional. We return to this point below.

We have seen how rightward movement of the subject in French can operate to govern INFL. The second strategy noted above is also available. INFL itself may move to a governed position. This is the process known as Complex Inversion (Kayne (1972)). Assume that the following movement is available:

- (17) INFL may move to v] (Chomsky-adjunction)

Such a movement will place INFL in a position where it is governed by V. In a matrix sentence, the trace of INFL will, properly, remain ungoverned, since government from a filled COMP will be blocked by the subject NP. This movement will derive structures like (18a). (We will explain the subject clitic il immediately below.)⁹



Notice that if a structure like (18a) is embedded, then we correctly predict an ungrammatical result:

- (18b). *je ne sais pas quand Jean est-il arrivé

In (18b), the trace of INFL will be governed from above by percolation, principle B(iv), since its maximal projection, $\bar{S}$, is governed by the matrix verb savoir, thereby violating D. Thus, the matrix/subordinate asymmetry in structures generated by Complex Inversion is predicted for French, exactly as the parallel V/2 asymmetry was predicted for German. We will shortly see that this analysis of Complex Inversion provides even more striking advantages.

Let us now add to our theory some principles to explain the presence and position of subject clitics in French. As noted by Kayne (1975), these pronouns cannot be separated from the tensed verb, by intervening material or by movement rules:

- (19)a. *il, je pense, est malade 'he, I think, is sick'
 b. *quand est arrivé (-t-) il?

Kayne suggests that these clitics are attached to the node V. We note that they may occur on either side of the verb, but not on both:

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- (20)a. il est malade
 b. est-il malade?
 c. *il est-il malade?

Let us assume that the subject clitic (SCL) is generated as a nominal in a special slot in V, whose order with respect to the main verbal material is unspecified, e.g.:

- (21) V { [_NSCL], $\bar{V}$ }

Further note that SCLs are nominative. As is well-known, they cannot occur as subjects of untensed sentences. Within the framework of Chomsky (1980, 1981) and other work, nominative Case assignment is a property of a tensed INFL. Let us assume this involves the transfer (or movement, see Stowell (1980)) of [+Case]-features from INFL to a unique eligible nominal. Let us further assume the following:

- (22) INFL is base-generated with nominative Case features. These must be assigned to an adjacent nominal.^{10,11}

When INFL remains adjacent to a lexical subject, Case-assignment is straightforward. The subject receives nominative Case, satisfying (23). No subject clitic will be generated, since nominative Case cannot be assigned twice by the same governor (see below). Caseless subject clitics will be ruled out by the filter prohibiting Caseless nominals (the Case filter of Chomsky (1980) and Rouveret & Vergnaud (1980)). Thus we will easily account for structures like:

- (23) [_{+Case}^{Jean}] [_{-Case}^{INFL}] [_{VP} [_{SCL}^e]-est parti] / Jean est parti
 'Jean has left'

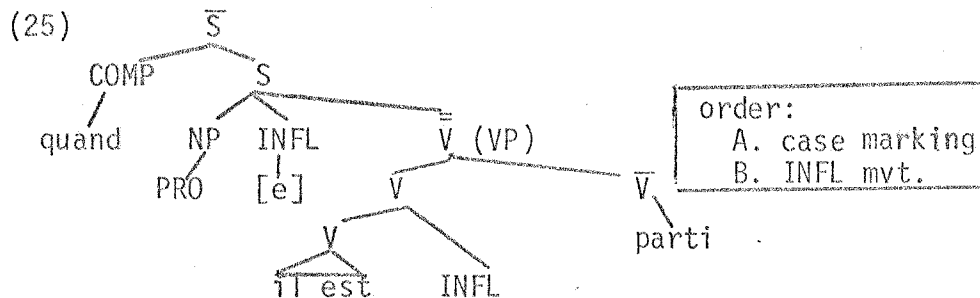
Lexical insertion is optional, although independent principles will govern the distribution of empty nodes, where lexical material is not inserted. When no material is inserted into subject position, we will call this NP "PRO", following Jaeggli (1980; forthcoming), who presents a theory of such governed PROs.¹² Assuming that Case must be assigned to lexical material, the absence of a lexical subject, combined with the requirements of (22), will force the generation of a SCL, in order to receive nominative Case. When INFL remains in its base-generated position, to the left of V, the adjacency requirement on nominative Case assignment will yield an SCL also to the left of V. Thus:

- (24) PRO [_{-Case}^{INFL}] [_{VP} [_{SCL}^{+Case}^{il}]-est parti] / il est parti

Occurrences of SCLs to the right of the tensed verb in Complex Inversion constructions like (18a) require a bit more comment. Let us reconsider cases of Stylistic Inversion, as in (16a-b). To satisfy the Case filter, the NP moved to the right must receive Case at some point in the derivation. Let us suppose it receives nominative Case, although there is little clear evidence on this point. To account for this Case-marking, two possibilities suggest themselves. (a) We might claim that Case marking can apply before movement, and that Case so assigned remains with the moved

NP. Alternatively, (b) Case marking might apply uniquely at S-structure (surface structure in the sense of Chomsky & Lasnik (1977)). In this case, nominative Case will be assigned, not directly to the moved NP in Stylistic Inversion constructions, but to its trace. We would assume that Case marking is transmitted from a trace to its controller. This latter theory is favored, in a different form, by Chomsky (1981), and we will show that our theory too must assume (b). Note that there is no a priori reason to prefer one theory over the other, since both are simple and do not appear to differ in the class of grammars permitted. Our reasons arise from consideration of Complex Inversion.

Let us return to cases of Complex Inversion like (18). Recall that these structures result from INFL moving to a position right-adjoined to V, as described in (17), and that this movement satisfies principle C by permitting INFL to be governed. Our theory of SCLs will automatically explain the occurrence of an SCL to the right of the verb. INFL will obligatorily jettison nominative Case features to an adjacent eligible nominal, which in this case will be a right-hand SCL. That movement is to the right follows from general principles of rightward movement. We must, however, prevent nominative Case marking from taking place before movement. This would yield an SCL to the left of V (cf. note 8):



*quand il est parti

Clearly this result must be blocked. We therefore assume theory (b) discussed below, and order Case marking after movement rules, at S-structure.

Let us return to Stylistic Inversion. Recall that we have claimed that INFL uniquely assigns nominative Case to an eligible nominal. In other words, INFL cannot assign nominative Case to two nominals. Thus, examples like (26) are straightforwardly excluded (ruling out the "right dislocation" reading).

(26) *quand elle est partie Marie?

Now compare (26) with examples of Complex Inversion like:

(27) quand Marie est-elle partie?

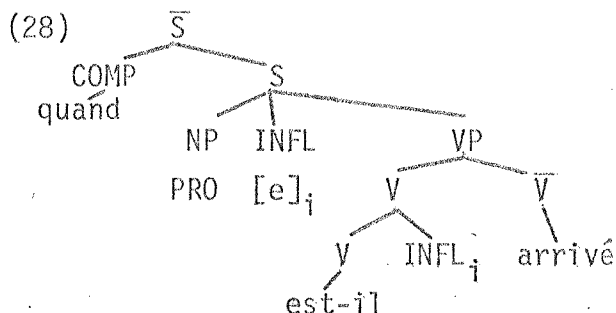
(27) appears to violate our principle, showing two nominative nominals. Nonetheless, this difference between Stylistic Inversion and Complex Inversion can be explained by a minor adjustment to our

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theory. Note that no movement of INFL is involved in the Stylistic Inversion construction, while INFL is assumed to move to V in Complex Inversion. Thus, in the latter case, but not in the former, S-structure will contain two occurrences of the node INFL -- the moved node and its trace. We know that the moved node can and must assign nominative Case, which can be realized on an SCL to the right of the verb. Suppose that the trace of INFL may also retain Case-marking properties, as an option. If the option is taken, then nominative Case can and must be realized on two nominals in S; the moved INFL will assign Case to a lexical NP, yielding structures like (27).¹³

Let us summarize our treatment of Case and SCLs so far. In a simple declarative there is only one INFL node; thus, either Case is assigned to an SCL realized to the left of the verb (to satisfy adjacency) and the subject is PRO, or else Case is assigned to a lexical NP (or its trace, as in Stylistic Inversion), and no SCL can be realized. When the INFL node is doubled in Complex Inversion, the moved INFL must jettison Case onto an SCL realized to the right of the verb. The trace of INFL in Complex Inversion is then another, optional, source of Case, permitting either a lexical subject or PRO, depending on the option taken. Thus, our theory correctly predicts the distribution of lexical subjects and subject clitics in French.

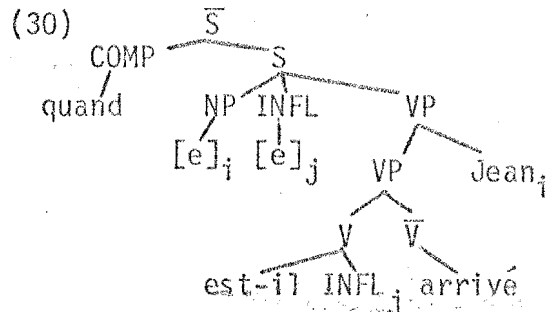
Now let us consider the occurrence of PRO in Complex Inversion structures. Notice that PRO, unlike trace, must crucially block government of INFL from COMP:



If PRO were transparent for adjacency, then lexically filled COMP (quand) would govern INFL trace, violating D.

This simple distinction between PRO and trace in subject position explains two otherwise puzzling restrictions on the application of Complex Inversion. First, if both Stylistic Inversion and Complex Inversion apply to the same clause, the result is ungrammatical, as in (29) (diagrammed in (30)).

(29) *quand est-il arrivé Jean?

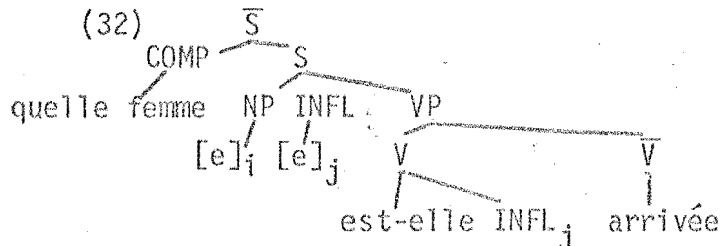


In (30), D is violated, because lexically filled COMP governs across the subject trace left by Stylistic Inversion.

The second restriction on Complex Inversion is that it is not permitted when the matrix subject is questioned:

- (31) *quelle femme est-elle arrivée?
'which woman is she-SCL arrived'

Questioning the subject leaves a trace in subject position:



Thus the trace of INFL is governed from lexically filled COMP in its base position. Of course, we correctly predict that the same sentence is grammatical without Complex Inversion:

- (33) quelle femme est arrivée?

The fact that principle D makes the correct predictions for (29), (31) and (33) is highly significant, in that it shows that D is a principle more general than any stipulation or filter which might simply ban inversion in subordinate clauses, even with an ad hoc exemption for Stylistic Inversion.

Thus, our theory of subject clitics and Inversion in French not only correctly predicts the distribution of lexical subjects, PRO subjects and SCLs, but it also predicts the matrix/subordinate asymmetry in the distribution of Complex Inversion, the lack of such an asymmetry for Stylistic Inversion, and, finally, the two curious restrictions on Complex Inversion just explained -- all by means of the same principles we used to explain V/2 in German plus an independently necessary theory of clitics.

4. Deriving the Principles

Let us now assume that our inflection-government theory is correct, and turn our attention to principles C and D. Why should these principles exist? Why are they virtually inverses of each other? Why do they refer to the node INFL? Finally, can these two principles be reduced to a single, more natural condition?

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We reason as follows. In every case where inversion is ruled out, either more than one INFL node is governed, or no INFL node is governed. This suggests that one and only one INFL node may be governed. We express this restriction by the following condition:

(34) Head Uniqueness Principle (HUP)

$\bar{S}$ must have one and only one governed head.

One might think of this principle as a sort of $\bar{X}$ -convention operating at LF (cf. Pesetsky (1981)). Suppose, for example, that a head is "visible" at LF only if it is governed. Perhaps categories with more than one such head would then be "incoherent", in some sense. Whatever deeper principles the HUP might reflect, it duplicates the effects of C and D by means of a single principle. In every case, if a sentence is ruled out by D, it turns out that there are two governed heads. In every case where C is crucial, there is no governed head at all.

One can imagine, however, a case where INFL might be governed in its base position, but then moves to an ungoverned position. This state of affairs would be ruled out by both C and D because these two filters distinguish lexical INFL from INFL trace, and neither INFL nor its trace would meet the filter that applies to it; this situation is allowed by the HUP, since only one head is governed. Are such cases attested? We think not. Thus, it seems to us that it is unnecessary to extend the HUP to apply to this imaginary case, since other principles will rule out such structures in every instance we can think of. For example, in French, the adjacency requirement on nominative Case assignment drastically limits the possible landing sites for INFL -- limits them, in fact, to a position adjacent to the tensed verb in VP, if INFL is not already governed in place. It then follows that German (and English) will lack Complex Inversion, since these languages do not have phenomena parallel to SCLs. We predict, therefore, that the only available governor for INFL in German is lexically filled COMP. In French, however, where government by percolation or from COMP is impossible the only possible governor is V. In short, the HUP seems to be no stronger than it has to be, and thus, by reducing redundancy, accounts more elegantly for the same data that C and D explain.

If the HUP is correct, then one naturally might suppose that it has other effects, especially if we can extend its application to all maximal categories. This remains a topic for future research.

Thus, our principles of government as expressed in B, combined with the HUP and our treatment of Case and subject clitics, provide a unified explanation for inversion and V/2 phenomena in French and German, respectively, and also account for the distribution of lexical subjects and subject clitics in French. No other theory derives this complex of results. We conclude that our findings supply convincing evidence that our notion of government, or a principle very much like it, plays a central role in the structure of universal grammar.

Footnotes

*We are indebted to Noam Chomsky, Jacqueline Gueron, Richard Kayne, Robert May, Jean-Yves Pollock, Alain Rouveret, Barry Schein and Donca Steriade for useful discussion of these ideas.

¹This was originally proposed by Hale (MIT lectures, 1977). See Marantz (1979) and references therein for some differing views.

²Note that this absorption creates a non-branching structure, so that INFL and V form a single node. Thus, V does not govern INFL in this structure, since, in a sense, it is INFL. Therefore, rule (3) alone cannot satisfy principle C.

³It need not be stipulated that (3) applies in D-structure. This is because any rule moving INFL will destroy the environment for (3). Since (3) is effectively obligatory (see text), it must apply before all such movement rules -- which is the desired result.

⁴The assumption that this is a rule of Chomsky-adjunction is not crucial here.

⁵We assume for yes/no questions an abstract Q which "lexically fills" COMP, and can govern INFL:

[S [_{COMP} Q] [_S hat Hans das Buch dem Herrn gestern gegeben]]

'Has Hans given the book to the man yesterday?'

⁶We assume that the extraposed S is originally generated under VP as a complement to V, and is subsequently moved to the right of the tensed verb (INFL) by a stylistic rule. Thus, at S-structure gesagt governs the S before it is moved to the right. Alternately, one might suggest that S is extraposed by Move α and adjoined to S. By our definition of government, the trace of INFL would then govern the extraposed S.

⁷Certain verbs allow V/2 in their complement clauses -- for example, sagen 'say'. We assume that these verbs do not (obligatorily) govern their S-complements. This similarly obviates the need for a separate V/2 constraint here, as proposed by Thiersch (1978).

⁸In colloquial French, (14) is grammatical for some speakers, who also accept Ou tu va? 'where (are) you go(ing)'. Such sentences are best with just a subject clitic, and sound worse when the subject is phonologically "heavy": *Ou Maximilien va? Other heaviness factors also seem to limit such uninverted questions, but we shall not discuss this matter further, nor shall we attempt to extend our analysis to these phenomena here.

We shall also pass over certain deviant instances of Stylistic Inversion:

??je ne sais pas pourquoi est parti Jean.

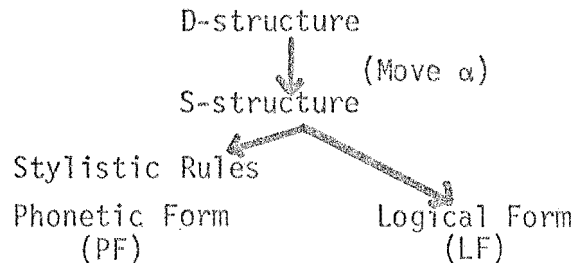
'I don't know why has left Jean'

*est parti Jean?

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See Kayne (1980b) for an independent analysis of these cases not directly relevant to our discussion.

⁹We assume a model of grammar like that of Chomsky & Lasnik (1977), as modified in Chomsky (1981):



We assume further that in French, unlike in German, the rule collapsing V with an adjacent INFL applies in PF and not before. This explains why French has no instance of Move α fronting INFL or a verb. These rules would have the effect of stranding the bound morpheme INFL, a circumstance ruled out on general grounds (see text and footnote 3). Thus a rule of subject-AUX inversion in French (**Où est Jean allé?* 'Where did Jean go?') is ruled out.

¹⁰The adjacency requirement, while not French-specific, must clearly be generalized to include languages where string-adjacency is not immediately relevant. German is such a language, as are so-called "non-configurational languages" (Hale (1979)).

¹¹This condition might be interpretable as a surface filter:

S-structure: * $\left[\begin{smallmatrix} \text{INFL} \\ +\text{Case} \end{smallmatrix} \right]$

¹²Alternately, we might adapt another idea of Jaeggli's and assume that the obligatory presence of an SCL in such sentences 'absorbs' verbal government in some sense, leaving PRO ungoverned. Other theories are also available: the empty subject position might be coindexed with the subject clitic, along the lines of Borer (1980), and will thus be properly governed, act like trace, and fall within the general theory. This is an area of ongoing research, and, while various proposals are available, the correct solution is not yet clear.

¹³Recall that the trace of INFL cannot assign Case to another, left-hand SCL, since by (21) a verb may have only a single SCL position, and the right-hand position is obligatorily present when INFL has moved into V.

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